

1. Agentic, Sequential & Decision-Driven Architectures

These frameworks move away from static, single-shot "retrieve-and-concatenate" workflows, reframing retrieval as an autonomous, sequential decision-making process managed directly by the model.

- **SoK: Agentic RAG (Finite-Horizon POMDP)**
- **Core Mechanics:** This framework formalises the open-domain agentic retrieval-generation loop as a **finite-horizon Partially Observable Markov Decision Process (POMDP)** 1. It explicitly models control policies and state transitions rather than relying on predefined, heuristic rules 1.
- **Uniqueness:** It provides the first unified modular decomposition that systematically categorises autonomous RAG systems by their planning mechanisms, memory paradigms, and tool-invocation behaviours 1. Crucially, it analyses the compounding systemic risks unique to agentic loops—such as **hallucination propagation, memory poisoning, retrieval misalignment, and cascading tool failures**—providing a definitive evaluation roadmap 1.
- **Graph-O1 (Monte Carlo Tree Search with Reinforcement Learning)**
- **Core Mechanics:** Graph-O1 frames graph reasoning as a multi-turn interaction between an LLM agent and a text-attributed graph environment 2. It integrates **Monte Carlo Tree Search (MCTS)** with end-to-end reinforcement learning 2.
- **Uniqueness:** Traditional GraphRAG serialises entire subgraphs into long text prompts, which rapidly overloads context limits 2. Graph-O1 solves this by using a unified reward mechanism trained via reinforcement learning, allowing the model to **selectively explore and retrieve only the most informative subgraph components** step-by-step 2.
- **A-RAG (Hierarchical Retrieval Interfaces)**
- **Core Mechanics:** A-RAG shifts the burden of retrieval planning directly to the agent's active reasoning loop by exposing multi-granularity retrieval interfaces 3.
- **Uniqueness:** Rather than retrieving in a single shot or using fixed pipelines, A-RAG provides the agent with **three granular, hierarchical tools: keyword_search, semantic_search, and chunk_read** 3. This progressive information disclosure allows the agent to dynamically adapt its retrieval depth, matching or exceeding state-of-the-art accuracy while dramatically reducing the volume of retrieved tokens 3.

2. Dynamic, Verification-Centric & Selective Abstention Architectures

These architectures determine exactly *when* to retrieve, *what* to trust, and *how* to verify that retrieved evidence is sufficient to justify an answer, enabling safe selective abstention.

- **SURE-RAG (Evidence Sufficiency Verification)**
- **Core Mechanics:** SURE-RAG frames selective answering as a formal evidence sufficiency verification problem 4. It uses a shared pair-level claim-evidence verifier to calculate local relation distributions 4.
- **Uniqueness:** Unlike independent passage scoring, SURE-RAG treats sufficiency as a **set-level property** 4. It aggregates local scores into five highly interpretable set-level metrics: **coverage, relation strength, disagreement, conflict, and retrieval uncertainty** 4. It yields an auditable selective score and a three-way

decision (**SUPPORTED, REFUTED, or INSUFFICIENT**), dropping unsafe, ungrounded answers by 37% through selective abstention 4.

- **QuCo-RAG (Pre-training Corpus-Grounded Uncertainty)**
- **Core Mechanics:** QuCo-RAG is a dynamic RAG framework that quantifies model uncertainty by analyzing objective statistics from the pre-training corpus 5.
- **Uniqueness:** Traditional dynamic RAG relies on model-internal signals (like logits or entropy), which are fundamentally unreliable because LLMs are poorly calibrated and often confident in errors 5. QuCo-RAG shifts to **objective pre-training statistics using Infini-gram** to query trillions of tokens in milliseconds 5. It flags low-frequency entities before generation and verifies entity co-occurrence during generation, triggering retrieval dynamically when objective co-occurrence drops to zero 5.
- **Predictive Prefetching (Asynchronous Latency Mitigation)**
- **Core Mechanics:** This framework decouples the synchronous bottleneck of RAG by executing predictive, asynchronous retrieval aligned with the model's decoding dynamics 6.
- **Uniqueness:** It utilizes three specialized components—a **retrieval predictor, a context monitor, and a query generator** 6. By exploiting semantic precursors that emerge in generation dynamics several tokens before uncertainty becomes critical, it **predicts when and what to retrieve** in advance 6. This maintains answer quality while achieving up to a 43.5% reduction in end-to-end latency and a 62.4% improvement in time-to-first-token 6.

3. Multimodal & Embodied Grounding Architectures

These systems extend RAG beyond pure text, mapping relational dependencies across visual regions, long video timelines, and physical manipulation environments.

- **MG²-RAG (Multi-Granularity Multimodal Graph)**
- **Core Mechanics:** MG²-RAG constructs a hierarchical multimodal knowledge graph by combining lightweight textual parsing with entity-driven visual grounding 7.
- **Uniqueness:** Unlike traditional multimodal graphs that rely on expensive "translation-to-text" pipelines, MG²-RAG **fuses textual entities and visual regions directly into unified multimodal nodes** 7. A multi-granularity graph retrieval mechanism aggregates dense similarities and propagates relevance across the graph, preserving atomic visual evidence and achieving up to a 43.3× speedup in construction overhead 7.
- **SignRAG (Zero-Shot Visual Retrieval)**
- **Core Mechanics:** SignRAG adapts the RAG paradigm to address zero-shot, fine-grained visual classification at scale 8.
- **Uniqueness:** Instead of training a massive, closed-world classifier, it employs a **hybrid pipeline**: a Vision Language Model (VLM) generates a rich textual description of an input image, which is then used to retrieve candidate matches from a vector database of reference designs 8. A Large Language Model (LLM) then reasons over the candidates to make a final fine-grained recognition, achieving high accuracy without task-specific training 8.
- **Affordance RAG (Embodied Robotics Memory)**
- **Core Mechanics:** Affordance RAG is designed for open-vocabulary mobile manipulation, enabling robots to execute natural language instructions over physical objects in large-scale indoor environments 9.

- **Uniqueness:** It constructs an **Affordance-Aware Embodied Memory** from pre-explored images 9. The framework retrieves candidate targets based on regional and visual semantics and **reranks them with physical affordance scores**, ensuring that the selected manipulation targets are highly likely to be executable in the real-world environment 9.
- **TV-RAG (Temporal-Aware Video RAG)**
- **Core Mechanics:** TV-RAG is a training-free architecture that couples temporal alignment with entropy-guided semantics to handle long-video retrieval and reasoning 10.
- **Uniqueness:** Standard visual RAG ignores temporal relationships across visual, audio, and subtitle channels 10. TV-RAG resolves this through a **time-decay retrieval module** that injects explicit temporal offsets into similarity calculations 10. It pairs this with an **entropy-weighted key-frame sampler** to select evenly spaced, highly informative frames, maximizing representativeness while minimizing token redundancy 10.

4. Structural & Tabular Reasoning Architectures

These frameworks address the structural breakdown that occurs when flat vector retrieval is applied to highly structured, multi-table databases.

- **FT-RAG (Fine-Grained Table Reasoning)**
- **Core Mechanics:** FT-RAG is a fine-grained reasoning framework designed for heterogeneous documents containing complex mixed tables and text 11.
- **Uniqueness:** Traditional table RAG retrieves coarse, whole-table chunks, losing semantic alignment 11. FT-RAG **decomposes tables into entry-level semantic units** to construct a structured graph 11. During retrieval, it uses a **structural neighbor expansion mechanism** to dynamically discover semantically connected entities across tables, followed by multimodal fusion to consolidate table context 11.

5. Efficient, Edge-Optimised & Hybrid Architectures

These systems optimize the trade-offs between computational budget, token efficiency, and noise-tolerant retrieval in real-world deployments.

- **HiFi-RAG (Hierarchical Filtering & Model Cascades)**
- **Core Mechanics:** HiFi-RAG is a hierarchical content filtering and two-pass generation pipeline designed to maximize open-domain precision and cost-efficiency 12.
- **Uniqueness:** It implements a **model cascade** 12. It leverages the cheap, high-speed **Gemini 2.5 Flash** to perform heavy pre-retrieval tasks: query formulation, hierarchical content filtering (filtering out 60.5% of irrelevant chunks by scoring webpage previews and snippets), and citation attribution 12. It reserves the heavy reasoning capabilities of **Gemini 2.5 Pro** strictly for the final pass, ensuring the context window is dense with high-signal evidence 12.
- **FD-RAG (Federated Dual-System RAG)**
- **Core Mechanics:** FD-RAG decouples lightweight memory access (System 1) from deliberative LLM reasoning (System 2) for edge and decentralized environments 13.
- **Uniqueness:** It learns a **semantic-aware adaptive hypergraph** over local corpora, distilling it into compact, edge-deployable Question-Answer (QA) memories 13.

Well-covered queries are answered instantly via direct memory matching 13. For complex, unrepresented queries, it invokes LLM reasoning over hypergraph evidence 13. To resolve cross-device fragmentation, it safely **aggregates anonymised memories across edge devices** without exposing raw documents 13.

- **Historical Multilingual RAG (OCR & Noise Tolerance)**
- **Core Mechanics:** Specifically designed for large-scale digital archives characterized by OCR corruption, spelling variations, and language drift 14.
- **Uniqueness:** It integrates **semantic query expansion and multi-query fusion via Reciprocal Rank Fusion (RRF)** to bypass vocabulary mismatches 14. By combining dense semantic and sparse lexical search pipelines, RRF acts as a smoothing mechanism to stabilize performance variance across corrupted texts and multilingual queries 14.

6. Security & Self-Improving Architectures

These frameworks introduce active defense mechanisms against corpus poisoning and enable safe, validated write-back loops that allow the database to evolve.

- **Bidirectional RAG (Safe Self-Improving Write-Back)**
- **Core Mechanics:** Unlike conventional, read-only RAG, Bidirectional RAG implements a validated feedback loop that enables safe corpus expansion using model-generated outputs 15.
- **Uniqueness:** To prevent "hallucination pollution" (where generated errors corrupt the corpus over time), it deploys a strict, **multi-stage acceptance layer** 15. A candidate document must pass **grounding verification (NLI-based entailment), rigorous attribution checking, and novelty detection** before being written back into the retrieval database, doubling standard corpus coverage 15.
- **RAGPart & RAGMask (Corpus Poisoning Defenses)**
- **Core Mechanics:** These represent lightweight, retrieval-stage defenses designed to mitigate the threat of corpus poisoning, where malicious documents are injected to hijack model outputs 16.
- **Uniqueness:**
- **RAGPart:** Exploits dense retriever training dynamics by **partitioning document sub-corpora** to neutralize the visual or textual influence of poisoned points 16.
- **RAGMask:** Actively **identifies suspicious tokens** by analyzing significant, unexpected similarity shifts that occur under targeted token masking, protecting the generation model without requiring costly retraining 16.

 Would you like me to generate a clean, step-by-step ASCII flow diagram comparing how **SURE-RAG**'s verification layer filters candidate answers versus **HiFi-RAG**'s hierarchical model cascade?